Chapter 20

Non-Carious Lesions and Their Management

"To be conscious that you are ignorant is a great step to knowledge".

—Benjamin Disraeli, 1845

The primary cause of tooth substance loss is dental caries; however, there are certain other non-carious conditions, which eventually result in loss of tooth structure. The cervical area, which is located in the gingival one-third of the facial and lingual tooth surfaces, is an area that exhibits unique clinical characteristics. The restoration of this region is challenging and the dentist has to be aware of the histopathogenesis of the various lesions occurring in this region. This chapter will present the various non-carious causes of tooth loss and their management strategies.

Non-Carious Lesions

Erosion

Definition. *Erosion* is the wear or loss of tooth surface by chemical action in the continued presence of demineralizing agents with low pH (Fig. 20.1A–C).

Classification Based on Etiology

Intrinsic erosion

This is a form of erosion caused due to endogenous acids of gastric origin. According to Scheutzel,[1] the following are the common aetiological causes of this kind of erosion.

I. Recurrent vomiting
   i. Eating disorders
      – *Anorexia nervosa*: This disorder is associated with extreme dietary restriction and profound weight loss.
      – *Bulimia nervosa*: This disorder is associated with repeated episodes of binge eating and self-induced vomiting.

   These personality disorders are considered to be a major cause of dental erosion due to chronic vomiting. They are most commonly found in young women in the age of 20–30 years.
   ii. Medical conditions
      – *Gastrointestinal disorders*
         - Peptic ulcer
         - Hiatus hernia
         - Intestinal obstructions
      – *Metabolic and endocrine disorders*
      – *Neurological disorders*
         - Side effects of drugs
         - Psychogenic vomiting syndrome
         - Chronic alcoholism and binge drinking
         - Pregnancy-induced vomiting

II. GERD (Gastro Esophageal Reflux Disease)
   Regurgitation in this disease occurs without any nausea or abdominal contractions. Erosion occurs when the acid reflux passes into the pharynx and come into contact with the lingual surfaces of the teeth.

III. Rumination
   Rumination is a syndrome consisting of repetitive, effortless regurgitation of undigested food within minutes after a meal. This disorder is found in young infants.

Extrinsic erosion

This is a form of erosion associated with extrinsic factors and according to Zero,[2] the following are the common etiological causes of this kind of erosion:

I. *Occupational factors*: Professional wine tasters and professional swimmers
II. *Diet*: Citrus fruit juices, acidic beverages and carbonated beverages
III. *Medicaments*: Aspirin and ascorbic acid (vitamin C)
IV. *Lifestyle*

Clinical Features

- Although erosive agents are the predominant causative factors, it is thought that toothbrushing and/or other abrasive agents in the diet may accelerate the loss of tooth structure, which is generally referred to as *erosive*



Figure 20.1 Erosion. (A) Crescent-shaped defects on enamel facial surfaces caused by exogenous demineralizing agent (from sucking on lemons several years previous to the time of the photograph). (B) Generalized erosion caused by endogenous fluids. (C) Idiopathic erosion lesion at the dentinoenamel junction is hypothesized to be associated with abnormal occlusal force. (D) Wedge-shaped lesions caused by abrasion from toothbrush. (E) Generalized attrition caused by excessive functional or parafunctional mandibular movements. (F) Enamel craze lines.

tooth wear. It is necessary to document the severity of the tooth structure loss and the specific areas that have been affected.

- If the defect was present only on the palatal surfaces of upper teeth, the diagnosis would be different from finding erosion on the occlusal surfaces of the lower molars.
  - Regurgitation of stomach acid can cause this condition on the palatal surfaces of maxillary teeth (particularly anterior teeth).
  - Extrinsic erosion commonly leads to the dissolution of the facial aspects of anterior and buccal aspects of posterior teeth.
  - Erosion processes may also be involved in the loss of the tooth structure with a clinical presentation of 'cupped-out' areas on occlusal surfaces.
- Exogenous acidic agents such as lemon juice (through sucking on lemons) may cause crescent-shaped or dished defects (rounded as opposed to angular) on the surfaces of exposed teeth (Fig. 20.2).
- Endogenous acidic agents such as gastric fluids cause generalized erosion on the lingual, incisal and occlusal surfaces. The latter defective surfaces are associated with the binge–purge syndrome in bulimia, or with gastro-oesophageal reflux disease (GERD).
- Many patients with GERD are often not aware of their gastric symptoms or do not associate them with the problems with their teeth. Consultation with a physician to obtain a proper diagnosis of GERD may assist in the diagnosis and management of erosion.
- Other sources of erosion can be the use of sports drinks, herbal teas and vomiting associated with chemotherapy



Figure 20.2 Erosion lesions on the labial surfaces of the maxillary and mandibular anterior teeth.
(From V Gopikrishna: *Preclinical manual of conservative dentistry*, ed 2, 2014, Elsevier.)



Figure 20.3 Attrition in a mandibular molar with sharp cuspal margins.(Courtesy: Dr V Gopikrishna.)

and, in the case of alcoholism, the presence of stomach contents in the mouth during periods of excessive alcohol consumption.

- The *flow* and *buffering capacity of saliva* are important factors in chemical erosion when other factors are present.
- Understanding, identifying and diagnosing tooth damage secondary to erosion (erosive tooth wear) is essential if management of the disease process is to be successful.

Attrition

Definition. *Attrition* is the mechanical wear of the incisal or occlusal surface as a result of functional or parafunctional movements of the mandible (tooth-to-tooth contacts) (Fig. 20.1E). Attrition also includes proximal surface wear at the contact area because of physiologic tooth movement.

Etiology

If significant abnormal attrition is present, the patient's functional movements should be evaluated, and inquiry needs to be made about any habits creating this problem, such as tooth grinding, or *bruxism*, usually resulting from:

I. Stress
II. Airway issues
III. Sleep apnoea

Clinical Features

I. In some older patients, the enamel of the cusp tips (or incisal edges) is worn off, resulting in cupped-out areas because the exposed, softer dentin wears faster than the surrounding enamel.
II. Sometimes, these areas are an annoyance because of food retention or the presence of peripheral, ragged and sharp enamel edges (Fig. 20.3).
III. Heavy occlusal loading from clenching may result in the presence of 'craze lines' which are limited to enamel (i.e. do not progress through the DEJ into dentin; Fig. 20.1F). Craze lines are not sensitive and do not require treatment but may be evidence of excessive masticatory muscle activity.
IV. Slowing such wear by appropriate restorative treatment is indicated. The sharp edges can result in tongue or cheek biting; rounding these edges does not completely resolve the problem but does improve comfort.

Non-carious Cervical Lesions

Abrasion

Definition. *Abrasion* is the abnormal tooth surface loss resulting from direct forces of friction between teeth and external objects or from frictional forces between contacting teeth components in the presence of an abrasive medium.[3]

Etiology

Abrasion may occur from:

I. Improper brushing habit: Factors influencing the role of tooth brushing in abrasion are:
   i. Brushing technique
   ii. Brushing force
   iii. Bristle stiffness
   iv. Time
   v. Frequency

II. Habitual chewing on hard objects (e.g. paper clips, pens, pencils)
III. Chronic use of agents with high abrasivity (e.g. smokeless tobacco, inadequately washed vegetables)

Clinical Features

- The loss of tooth structure in the cervical areas (*abrasion*) is commonly seen as a rounded notch in the gingival portion of the facial aspects of teeth (Fig. 20.1D).
- The surface of the defect is usually smooth. The presence of such defects does not automatically warrant intervention. It is important to determine and eliminate the cause (Fig. 20.4).

Abfraction

Definition. Strong eccentric occlusal force resulting in microfractures at the cervical area of the tooth causing *wedge-shaped* defects is termed as *abfraction* (Fig. 20.5).[4]



Figure 20.4 Abrasion lesions in the cervical third of a maxillary canine and first premolar. (From V Gopikrishna: *Preclinical manual of conservative dentistry*, ed 2, 2014, Elsevier)

Etiology

Teeth are not rigid structures and undergo deformation (strain) during normal loading.[5] Intra-oral loads (forces) vary widely and have been reported to range from 10 N to 431 N, with a functional load of 70 N considered clinically normal.[6] The number of teeth, type of occlusion and occlusal habits of patients (e.g. bruxism) affect the load per tooth.

Abfraction is caused due to *tooth flexure* in patients with abnormal occlusal interactions. The mechanism is explained in Box 20.1.

Clinical Features

I. These lesions are characterized by *sharp notch* or *wedge-shaped lesions* instead of the saucer-shaped defects associated with other non-carious cervical lesions.
II. The maximal abfractive stresses generated are at the cervical area in the thinnest region of enamel at the cementoenamel junction (CEJ) (Figs. 20.6 and 20.7).[7–11]
III. These forces can also cause the *loss of bonded Class V restorations* in preparations with no retention grooves (Fig. 20.8).

Box 20.1 Mechanism of abfraction

Tooth flexure during abnormal occlusal interaction
↓
Lateral or axial bending of the tooth
↓
Tensile and compressive stresses generated in the cervical region
↓
Strain leading to microfractures in cervical enamel and tooth loss
↓
Notch shaped abfraction lesions

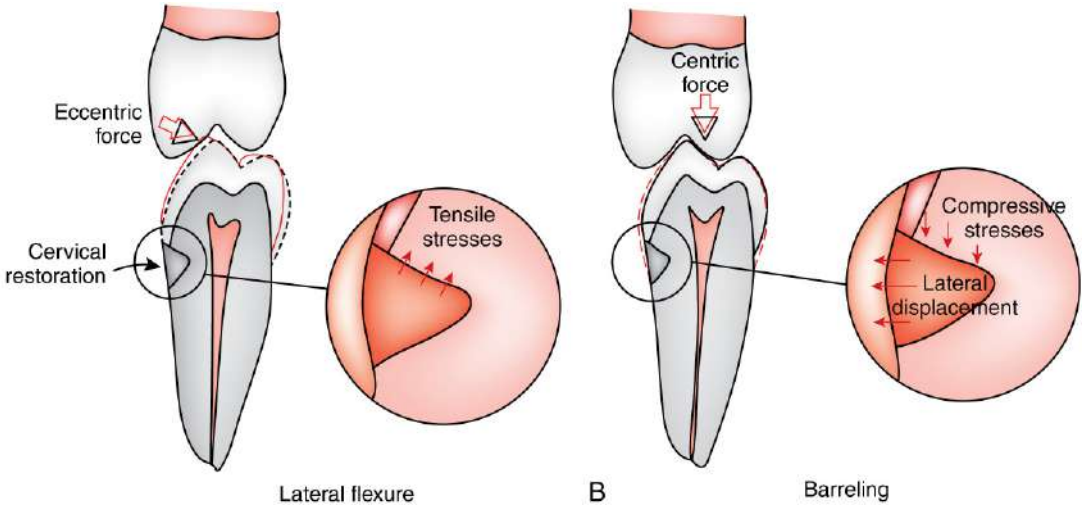


Figure 20.5 Class V lesions on two premolars suspected of being abfractions arising from tooth flexure. (From Grippo JO., et al.: Attrition, abrasion, corrosion, and abfraction revisited: a new perspective on tooth surface lesions, *J Am Dent Assoc* 135:1109–1118, 2004.)

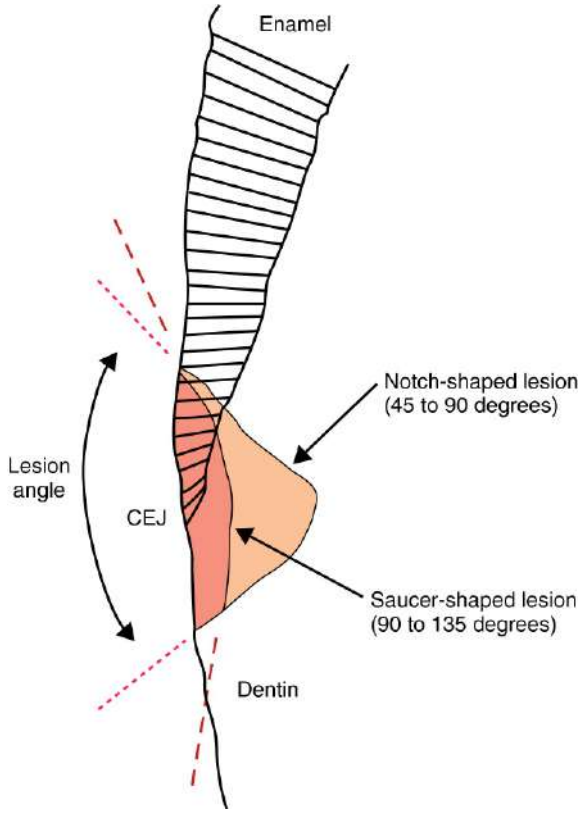


Figure 20.6 Schematic view of microfractures developing between enamel rods in cervical enamel. The enamel near the junction of the cementoenamel junction and dentinoenamel junction is prismless.

IV. Such fractures predispose enamel to loss when subjected to toothbrush abrasion and chemical erosion. This process may act as a key in the formation of some Class V defects. Additionally, in unbonded or leaking restorations, this flexure of dentin may produce changes in fluid flow and microleakage, leading to sensitivity and pulpal inflammation.

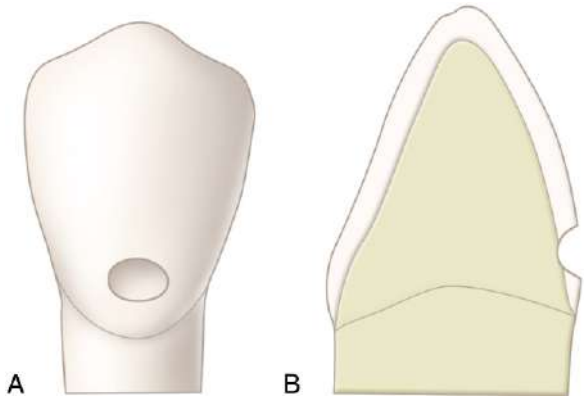


Figure 20.7 Schematic diagram of tooth flexure creating cervical stresses. (A) Lateral flexure results from eccentric forces that produce tensile stresses at the marginal interface with cervical restoration placed in the facial cementoenamel junction region. (B) Barrelling results from heavy centric forces that produce compressive stresses along the marginal interface with cervical restoration in the entire cementoenamel junction region, resulting in lateral displacement (loss) of the restoration.
(From Heymann HO., et al.: Tooth flexure effects on cervical restorations: a two-year study, *J Am Dent Assoc* 122:41–47, 1991.)

Treatment of Abrasion, Erosion, Abfraction and Attrition

The primary goal of management should be to halt or modify the etiology of the problem. Box 20.2 enumerates

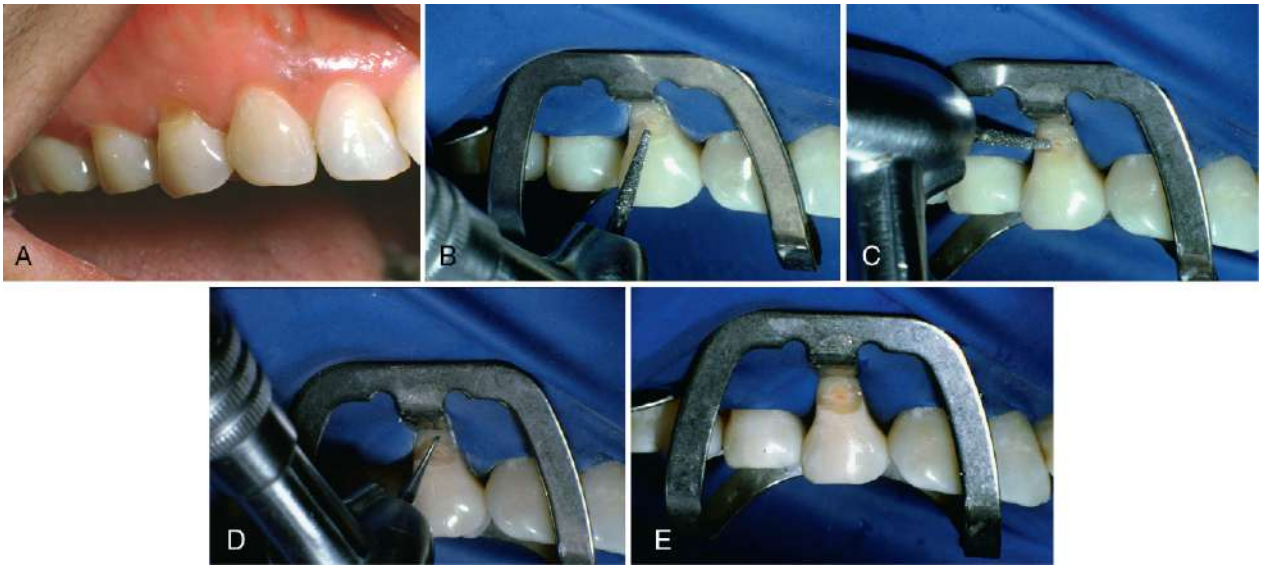


Figure 20.8 Class V tooth preparation for abrasion and erosion lesions. (A) Preoperative notched lesion. (B to D) Beveling the enamel margin and roughening the internal walls, (E) Completed preparation with etched enamel.
(From Bayne SC., et al.: Class V angulation, size and depth effects on composite retention [abstract 1669], *J Dent Res* 71A:314, 1992.)

Box 20.2 Considerations for restorative management of non-carious lesions

- The defect is sufficiently deep to compromise the structural integrity of the tooth
- Intolerable sensitivity exists and is unresponsive to conservative desensitizing measures
- The area is affected by caries
- The defect contributes to a periodontal problem
- The area is to be involved in the design of a removable partial denture
- The depth of the defect is judged to be close to the pulp
- The defect is actively progressing
- The patient desires esthetic improvements

the considerations required for initiating restorative management of these lesions.

- Areas of significant occlusal attrition that have exposed dentin are sensitive, or annoying should be considered for restoration or at least protection from additional loss of tooth structure.
- The general consensus among experts is that the etiology of non-carious cervical lesions (NCCLs). lesions is multifactorial and that well-designed clinical research studies are required to gain better understanding relative to the etiology of cervical abrasion and abfraction defects. The presence of such defects does not automatically warrant intervention.
- A treatment plan for definitive restorations must include an occlusal analysis (which requires articulated diagnostic models) as part of the comprehensive examination. Careful consideration of related information from the patient assessment and examination process is essential if all aspects of the etiology are to be identified and risk factors reduced.
- Also, occlusal guard therapy should be considered for nocturnal protection of indirect restorations completed as a part of the definitive phase. The creation of an occlusal guard, for nocturnal use, may be indicated with a diagnosis of sleep-related bruxism.

Clinical Performance of Adhesive Restorations in NCCL

Class V cervical tooth preparations, by definition, are located in the gingival one-third of the facial and lingual tooth surfaces. Because of esthetic considerations, *composites* are most frequently used for the restoration of Class V lesions in anterior and premolar teeth. The other alternative material used in certain clinical conditions is employing a *GIC restoration*. Several clinical factors may influence the success of an adhesive restoration. The mineral content of dentin increases in different situations, including:

- Aged dentin,
- Dentin beneath a carious lesion and
- Dentin exposed to the oral cavity in NCCLs in which the tubules become obliterated with tricalcium phosphate crystals.[12–14]

Role of Sclerotic Dentin

- The dentin that undergoes these compositional changes is called *sclerotic dentin* and is much more resistant to acid etching than 'normal' dentin.[15] Consequently, the penetration of a dentin adhesive is limited.[15–17]
- Irrespective of the use of a etch-and-rinse or a self-etch technique, *bonding to sclerotic dentin in NCCLs has resulted in low bond strengths*.[14,18] Additionally, the clinical effectiveness of dentin adhesives is less in sclerotic cervical lesions than in normal dentin.[19,20] Nevertheless, some specific dentin adhesives may perform better in sclerotic dentin than in normal dentin.[21]

Role of Masticatory Forces

- Some evidence suggests that masticatory forces not only might cause NCCLs but also might contribute to the failure of Class V restorations.[18,22,23] Bruxism or any other eccentric movement may generate lateral forces that cause concentration of stresses around the cervical area of the teeth.
- Although this stress may be of very low magnitude, the fatigue caused by cyclic stresses may cause failure of bonds between resin and dentin. Considering the challenges to successfully bond to NCCLs, these have been used often as ultimate clinical test to assess adhesives.

Role of Type of Composite

- The type of composite used might play an important role in clinical longevity of Class V, as well as Class II, restorations.[24] Composites shrink as they polymerize, but the amount of shrinkage depends on the inorganic load of each specific composite.
  - *Microfilled composites* have a low elastic (or Young's) modulus, which means that they are better able to relieve stresses caused by polymerization or by tooth flexure.[25,26] Materials that have a higher Young's modulus do not relieve stresses by flow; they are unable to compensate for the stresses accumulated during polymerization. These stresses subsequently might be transferred to the adhesive interface and cause debonding. As adhesives have improved however, restorative material stiffness might be less important. A 2-year clinical study of a three-step etch-and-rinse adhesive showed *no difference in retention rates of Class V composite restorations based on stiffness of the restorative material.*[27]

- Although GIC-based materials are usually outperformed by adhesively bonded restorations in in vitro studies, their clinical retention to tooth structure is *more effective* and *durable* than any other type of adhesive material.

Clinical Notes

The reader is referred to Chapter 19 and Figs 19.48–19.51 to see clinical techniques for using composites and glass ionomer materials in cervical Class V defects.

Role of Type of Adhesives

- A systematic review published in 2005[28] analyzed the literature published from January 1998 to May 2004 on the clinical effectiveness of adhesives to restore NCCLs to determine if simplified adhesives were as clinically effective as conventional three-step adhesives in terms of retention rate.
  - Three-step etch-and-rinse adhesives and two-step SEAs (Self Etch Adhesives) showed a clinically reliable clinical performance.
  - The clinical effectiveness of two-step etch-and-rinse adhesives was less favourable, while the worst clinical performance was found for one-step SEAs.
- A meta-analysis published in 2010, for which 50 clinical studies and 40 adhesive systems matched the inclusion criteria, reported that the clinical performance of dentin adhesives in NCCLs up to 3 years is significantly influenced by
  - *Type of adhesive system used*: GIC (Glass Ionomer Cement)-based materials, two-step SEAs and three-step etch-and-rinse adhesives performed better than adhesives of the other categories and
  - *Whether or not the dentin/enamel is roughened*.[29] Roughened dentin resulted in a statistically significantly higher retention rate than unprepared dentin.
- Peumans et al. (2014)[30] conducted a systematic review in which they analyzed annual failure rates in NCCLs in publications, up to 2013.[30] Dentin adhesive materials were divided into six main categories: three-step etch-and-rinse adhesives, two-step etch-and-rinse adhesives, two-step SEAs, one-step SEAs, GIC-based materials and self-adhesive composites. Both classes of SEAs were further subdivided as 'mild' and 'intermediately strong' (pH $\geq$1.5) and 'strong' (pH <1.5).[30]
  - GIC-based materials had a 2% annual failure rate (AFR), which was the lowest for all adhesive classes analyzed.
  - The AFR for two-step mild SEAs was 2.5% and for three-step etch-and-rinse adhesives was 3.1%. Higher AFRs were measured for one-step strong SEAs (5.4%), two-step etch-and rinse adhesives (5.8%) and two-step strong SEAs (7.9%).
  - Selective enamel etching did not significantly influence the retention rate of SEAs.
- Because universal adhesives are fairly recent materials, systematic reviews of clinical studies are not yet available. Two clinical trials have reported acceptable clinical effectiveness for the first universal adhesive introduced to the market, Scotchbond Universal adhesive (Single Bond Universal in some regions) (3M Oral Care).[31,32] A 3-year clinical study compared this adhesive applied in different adhesion strategies (self-etch, selective enamel etch, etch-and-rinse applied on moist dentin, etch-and-rinse applied on air-dried dentin) and concluded that while there was no statistical difference among bonding strategies, there were signs of marginal degradation when the universal adhesive was applied in full self-etch mode.[31]

Clinical Note

Results from recent clinical studies and systematic reviews suggest that:

- Chemical bonding to dentin calcium may play a crucial role in stable adhesion to NCCLs, as **GIC-based materials** and **mild two-step SEAs result in excellent retention rate**.
- *Roughening dentin* in NCCLs may increase retention rate, whereas enamel bevelling does not seem to be a significant factor.
- Although there is conflicting data on the influence of rubber-dam isolation, its use may decrease the failure rate of restorations in NCCLs.
- GIC-based materials are still the standard to restore NCCLs in terms of retention rate. Typical GIC-based restorations are shown before and after treatment in Fig. 20.9 and 20.10.

Non-carious Cervical Lesions and Mucogingival Surgery

Non-carious cervical lesions (NCCLs) are characterized by the loss of tooth structure at the cervical area of a tooth due to abfraction, abrasion and/or erosion (Fig. 20.11). These lesions are non-carious but may need to be managed provided one or more of these factors are present:

I. The patient is concerned with esthetics,
II. Substantial tooth sensitivity that does not resolve with other treatment options (i.e. desensitizing agents),
III. Lesion is progressing and there is risk of pulpal exposure or
IV. Lesion is progressing and the tooth integrity is at risk.

To deliver the best possible treatment preserving the natural contour and integrity of the gingival tissues around the affected tooth or teeth, options other than restoration



Figure 20.9 A Class V non-carious lesions in the mandibular incisors. B. Lesions restored with chemical cure Glass Ionomer restorations. (Courtesy: Dr Matteo Basso.)



Figure 20.10 A Class V non-carious cervical lesions in the mandibular canine and premolars. B Lesions restored with resin modified glass ionomer restorations.(Courtesy: Dr G Knight.)



Figure 20.11 Non-carious cervical lesions are characterized by the loss of tooth structure at the cervical area of a tooth and multifactorial etiology. Treatment may involve soft tissue grafting of the area depending on extent of lesion and periodontal condition with or without concomitant tooth restoration.

only are available and should be considered. In the event of gingival recession and/or lack of keratinized/attached gingiva with concomitant presence of a NCCL, careful evaluation of the site should be performed to determine the appropriate treatment:

I. Restoration
II. Soft tissue coverage (graft)
III. Combined approach

Predictable soft tissue root coverage of recessions with subcutaneous connective tissue graft can be accomplished with recessions that fall under the Miller classification of Class I and Class II.[33]

The Miller classification has been widely used to classify recessions based on the amount of tissue loss in relationship to the mucogingival junction, width of the tissue loss

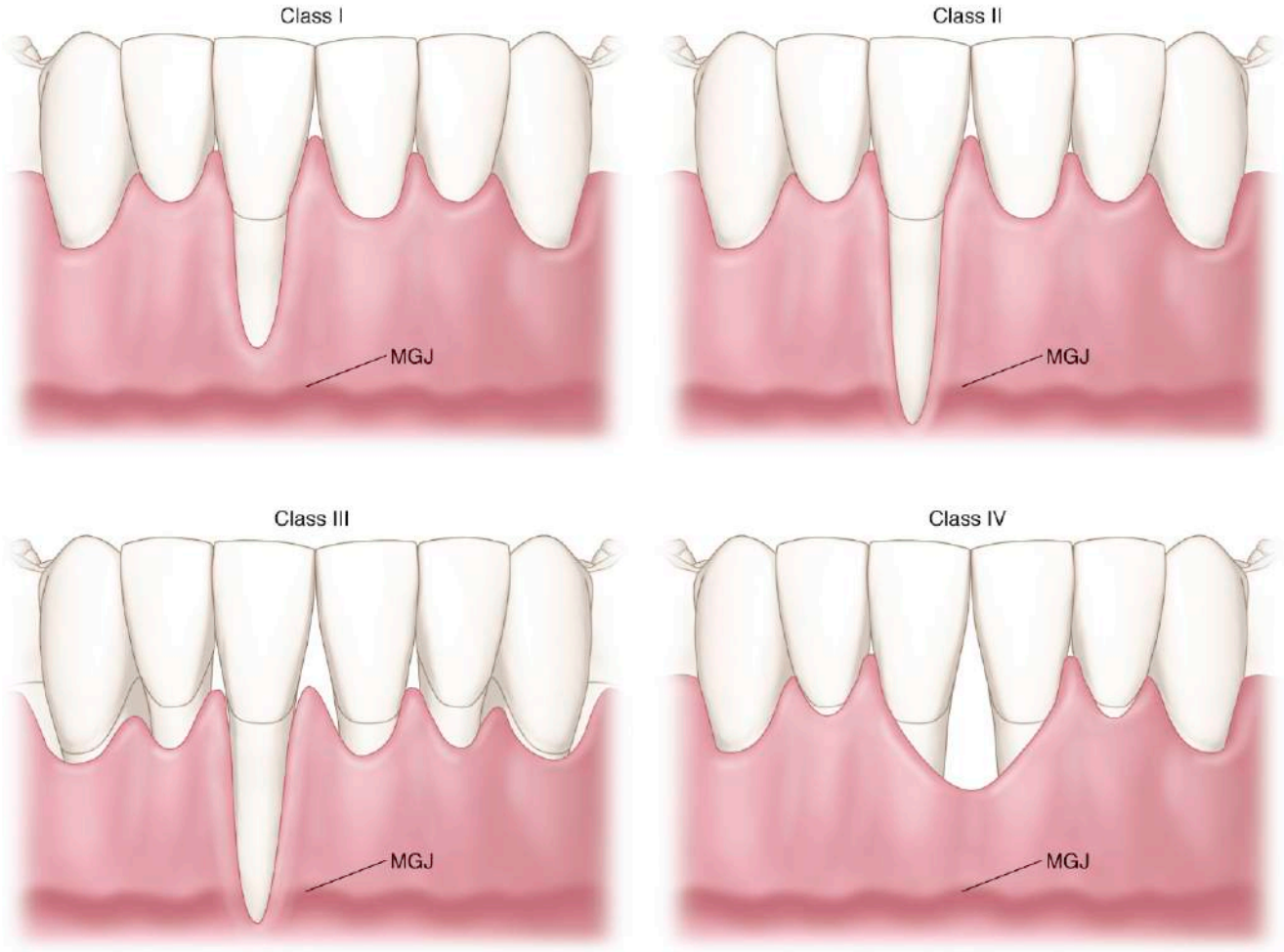


Figure 20.12 Miller's classification of gingival recessions. Class I: recession does not extend to the mucogingival junction (*MGJ*). Class II: recession extends to or beyond the MGJ; 100% of root coverage can be expected for Class I and II. Class III: recession extends to or beyond the MGJ with loss of interdental tissue or there is malpositioning leading to inability to cover 100% of the root surface. Class IV: recession extends to or beyond the MGJ where there is advanced loss of interdental tissues and root coverage is not anticipated.

around the facial or lingual aspect of the tooth and loss of soft and hard interproximal tissue[34] (Fig. 20.12):

I. Class I: Marginal tissue recession that does not extend to the mucogingival junction. There is no periodontal loss (bone or soft tissue) in the interdental area, and 100% root coverage can be anticipated.
II. Class II: Marginal tissue recession that extends to or beyond the mucogingival junction. There is no loss of interdental bone or soft tissue and 100% of root coverage can be expected.
III. Class III: Recession that extends to or beyond the mucogingival junction. There is loss of interdental bone and/or soft tissue or there is malpositioning of the teeth leading to inability to cover 100% of the root surface.
IV. Class IV: Marginal tissue recession that extends to or beyond the mucogingival junction where there is advanced loss of interdental tissues and root coverage is not anticipated.

After gingival grafting, full root coverage can usually be expected up to the *level of the interdental bone*. However, soft tissue coverage of a cervical lesion that is located mostly in enamel is not recommended.[35]

I. Indeed, no true tissue–tooth attachment is expected to happen between the grafted soft tissue and the tooth enamel to help stabilize the graft. Further, such location is typically at a far distance from the bone, which restricts vascular supply to support the grafted tissue in that area.
II. Clinical researchers have reported that NCCLs, just like interdental tissue loss, are a significant negative factor in the prognosis of gingival root coverage. The authors did not elaborate on the possible reason. It is possible that occlusion and hygiene factors are involved in the lower success of grafting of NCCLs.

The following rationale is followed when treating a NCCL.

- *Lesions that exceed approximately 2 mm in axial depth*:
  - These lesions typically need restoration. This will depend on progression rate, symptomatology and desire to restore.
  - Aside from diagnosing and treating the etiology, fast-developing lesions should be restored. Restoration also may be indicated for NCCLs that continue symptomatic after attempts to reduce sensitivity with desensitizing products.
  - Soft tissue grafting in addition to restoration may be recommended when there is lack of attached gingiva or a thin tissue biotype.[35]
- *Lesions whose axial depth is <2 mm*:
  - Grafting only is indicated in cases of limited tooth loss.
  - That is based on the need for grafting due to recession or lack of keratinized/attached gingiva. While Miller Class I and Class II cases have good prognosis, the long-term stability has been questioned. That said, the patient may opt for periodic application of desensitizing products in lieu of the surgical procedure.
  - A randomized clinical trial evaluating the outcome of connective tissue graft and resin-modified glass ionomer restorations for the treatment of gingival recession and NCCLs concluded that graft alone or graft with restoration showed increased attachment level and soft tissue coverage for up to 2 years in patients with Miller Class I recessions.

When considering mucogingival surgery:

I. The keratinized tissue and attached gingiva levels must be evaluated. There is a difference between attached and keratinized gingiva. Keratinized gingiva involves the attached and the free gingival tissue. Since the attached gingiva is bound to the underlying periosteum, the clear presence of keratinized gingiva is seen as protective of the periodontium against inflammation and bone loss. Recommendations have been made to increase the band of keratinized/attached gingiva prior to subgingival restorations that will be in close proximity with the gingival margin (or subgingivally after grafting). The presence of a narrow keratinized tissue band (i.e. less than 2 mm) has been linked to an increased chance of gingivitis.[25] However, others have found that gingival health can be maintained with little or no keratinized tissue in the absence of plaque accumulation and inflammation. Soft tissue grafting to increase the band of keratinized/attached gingiva may be recommended in patients with poor hygiene and restorations in close proximity to the gingival tissues.

II. Another indication for mucogingival surgery includes teeth that lack adequate band of keratinized/attached gingiva and that will undergo orthodontic therapy. Increase of the keratinized/attached gingiva band may be necessary to avoid gingival recession. This is particularly significant for patients with thin periodontium. Despite this, teeth should always be moved inside of their alveolar housing to minimize chances of recession. Recessions cannot be predictably covered when the position of the tooth in the arch is far outside the alveolar plate.

Summary

The management of non-carious lesions involves the twin approach of halting the etiological factors and then using the appropriate restorative protocol. These lesions can gradually cause substantial tooth loss and hence it is important that they are identified at the earliest and treated accordingly.

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